Seunghyun Lee
https://qwertyyuiop1234.github.io
Undergraduate Researcher

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Research Interests

- **AI Systems** - AI Performance Optimization, AI Hardware Accelerator
- **Models** - Large Language Models (LLM), World Model

Education

- **Konkuk University** Seoul, Korea
  B.S. in Computer Science and Engineering *2022 – Present*
  - Cumulative GPA: 4.24 / 4.5 Major GPA: 4.46 / 4.5

Publications

- **KV Cache Reuse in Korean Multi-Agent Systems: A Cross-Lingual Difference in Entropy-Gate Decisions**
  Seunghyun Lee, Jaehyung Seo
  Annual Conference on Human and Cognitive Language Technology (HCLT 2026) – **Oral Presentation**
  Related patent application pending.
  - Analyzed how KVCOMM-style KV cache sharing behaves on paired English–Korean HRM8K (Llama-3.1-8B-Instruct) and led problem formulation, implementation, experiments, and writing.
  - Found Korean reuses KV cache more often despite 1.34× more tokens, traced the gap to the entropy gate and to length-driven distance spread in layers 22–31, and showed it costs more accuracy.

Research Experience

- **ELION LAB, Konkuk University** Seoul, Korea
  Undergraduate Researcher (Advisor: Prof. Jaehyung Seo) *2026 – Present*
  - Researching KV cache reuse in multi-agent LLMs and presented the cross-lingual study as first author at HCLT 2026 (Oral).
  - Participating researcher in three funded projects: NRF Excellent Young Researcher Program, KOCCA, and NRF Basic Research Program.
  - **KV-cache-friendly skill schemas for skill-based agents** (ongoing)
    Reordering prompt segments to raise prefix-cache hit rate for SkillRL agents (Qwen2.5-7B-Instruct RL checkpoint) on ALFWorld. Found the action segment to be uniquely position-sensitive and now designing controlled experiments to verify it.

Projects

- **InfraTech-KR** Nov. 2026 – Present
  Personal study project
  - Translating the open AI-infrastructure curriculum *InfraTech* (Chinese) into Korean while studying LLM serving internals: vLLM/SGLang schedulers, KV cache management, prefix caching, and speculative decoding. Notes are published on my blog.
- **LLM from Scratch** Mar. 2026 – Present
  Personal project
  - Building a GPT-style language model from scratch in PyTorch without high-level libraries, currently implemented up to multi-head self-attention.
- **Bargate – AI consultation assistant for lawyers** Dec. 2025 – Apr. 2026
  Tech Lead
  - Led development as tech lead, built the backend, and integrated ElevenLabs speech-to-text for consultation transcription.
  - Passed the first round of the national startup program 모두의 창업.

- **Juchajang – AI vacant parking spot finder** 2025
  Tech Lead
  - Led a team building an AI service that guides drivers to vacant parking spots, implemented with FastAPI, React, and PostgreSQL.
  - Ranked in the top 20 of 550 teams at the 2025 K-Hackathon (Ministry of Science and ICT).

Awards & Honors

- **Academic Scholarship (70% Tuition Waiver)** 2026
  Konkuk University
  - Awarded for outstanding academic performance in the second semester of sophomore year.
- **Academic Scholarship (30% Tuition Waiver)** 2026
  Konkuk University
  - Awarded in the first semester of junior year for outstanding academic performance.
- **Dean's List (Two-Time Recipient)** 2026
  Konkuk University
  - Recognized twice for being within the top 5% of academic achievers.
- **Gwacheon Scholarship** 2026
  Gwacheon Aehyang Scholarship Association
  - Awarded for outstanding academic performance in the second semester of sophomore year.
- **K-Hackathon** 2025
  MSIT (Ministry of Science and ICT)
  - Ranked in the top 20 out of 550 teams nationwide.
- **Certificate of Appreciation (COA)** 2025
  1-23 Infantry Battalion, United States Army
  - Received for outstanding service and dedication to duty as a KATUSA Medic in the 7th Infantry Division.

Skills

- **Programming Languages** - Python, C++
- **Libraries & Frameworks** - PyTorch, NumPy, Pandas
- **Languages** - Korean (Native speaker), English (Conversational)

Military Service

- **Republic of Korea Army (KATUSA)** Camp Casey, Korea
  Combat Medic *2023 – 2025*
  - Served as a KATUSA (Korean Augmentation To the United States Army) medic.
  - Provided medical support and emergency care in a joint military environment.

Community

- **Member** 2025 – 2026
  Google Developer Groups on Campus (GDGoC)
  - Participated in a developer community hosted by Google, focusing on AI research and paper reviews.
  - Studied and analyzed research papers on AI topics including LLMs and Computer Vision.
  - Engaged in collaborative discussions and knowledge sharing with colleagues.

Certificates

- Google Developer Groups on Campus (GDGoC) Program Completion
- BOOSTRUN: 인공지능 전문가 특강 1기 (AI Expert Lecture Series, 1st Cohort)